PTPC: Prototype-Based Test-Time Prediction Calibration for Few-Shot Object Detection

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Abstract

Few-shot detectors can produce plausible post-NMS boxes whose confidence omits two relative cues: competition among novel classes and evidence that a base class better explains the region. We ask whether these omissions can be repaired through inference-time rescoring. We introduce Prototype-Based Test-Time Prediction Calibration (PTPC), which comprises two modules: a contextual novel-prototype rescorer, which pools frozen foundation-model patch tokens and compares the predicted label with all novel support prototypes, and an independently usable Base-Confusability Guard, which continuously attenuates scores according to excess base- versus-novel prototype evidence. We evaluate full PTPC on CNN-based detectors and apply the Guard independently to foundation-model detectors that provide compatible region features. On PASCAL VOC, PTPC improves mean novel AP50 over paired detector-only baselines by 11.8 points for DeFRCN and 5.6 points for MFDC; the corresponding gains on MS COCO are 2.4 and 2.5 AP points. For native FSOD-VFM, the Guard adds 1.91 VOC AP50 and 0.60 COCO AP and for No-Time-to-Train 1.58 VOC AP50/0.75 COCO AP on our paired No-Time reproduction. Improvements hold across all 42 evaluated detector–shot settings. With all Guard parameters selected on the other benchmark and frozen, the standalone Guard also improves AP, AP50, ECE, and Brier score across all 38 evaluated settings for FSOD-VFM and our No-Time-to-Train reproduction. Changing the encoder alone does not reproduce the full PTPC result; an absolute-similarity control stays at native AP50.

1. Introduction

Few-shot object detection (FSOD) adapts detectors to novel categories from few labeled examples. With few training examples for novel classes, a detector may confidently predict one class even when another class is a better match. As a result, some novel-class predictions become less reliable. And after non-maximum suppression, these scores determine detection ranking and threshold decisions.

We study whether relative evidence can improve the ranking of these detections at inference time. By keeping boxes and labels fixed, we isolate the effect of confidence rescoring on detection performance. But it also introduces a limitation as score changes cannot recover missed objects, poor localization, or incorrect labels.

DeFRCN's Prototypical Calibration Block (PCB) [24] combines detector confidence with only the assigned-class prototype similarity using RoIAlign features from a pretrained CNN. This unary comparison omits competing novel and base classes.

We introduce Prototype-Based Test-Time Prediction Calibration (PTPC), with two modules (Fig. 1). The *contextual novel-prototype* module pools full-image patch tokens from a frozen foundation encoder within each box, retaining contextual information. It compares the pooled descriptor with all novel support prototypes and fuses the predicted class's relative compatibility with detector confidence. The *Base-Confusability Guard* continuously attenuates scores according to relative base-versus-novel evidence.

Contributions.

1. We combine contextual novel-class competition with base-versus-novel evidence in full PTPC and a native-descriptor Guard interface.
2. Paired VOC and COCO evaluations show ranking gains on CNN detectors and foundation-model detectors, with all Guard parameters selected on the other dataset for the latter. Both native hosts improve in AP, AP50, ECE, and Brier score.
3. Controlled ablations examine the scoring mechanisms, while a fixed-output bound identifies remaining ranking headroom.

2. Related Work

Detector adaptation and post-hoc rescoring. Few-shot detectors learn novel categories through meta-learning, feature reweighting, or fine-tuning [4, 13, 29, 35]. Model Calibration [5] changes detector components and training objectives to reduce bias. Other methods use support examples to adapt classification or representations: TIDE [18] uses a dynamic classifier and cross-attention at test time; PDA [12] learns a projection and optional prototype-conditioned RoI alignment; PS-TTL [7] updates a teacher detector using prototype-derived soft labels. PTPC rescores detections after NMS. It builds on DeFRCN's Prototypical Calibration Block (PCB) [24], which combines detector confidence with assigned-class prototype similarity using RoIAlign features. PTPC extends this comparison to competing novel prototypes and adds a separate comparison against base classes.

Prototypes and foundation-model FSOD. A prototype is a representative feature vector for a class, often formed by averaging the features of its labeled examples. Prototypical Networks [26] use class prototypes for metric-based few-shot learning. In FSOD, prototypes are widely used for support-feature aggregation, class conditioning, and refinement [1, 16, 20, 31]. Foundation-model methods use pretrained representations in different ways: DE-ViT compares region features with class and background prototypes [38]; FM-FSOD uses foundation-model features for proposal classification [11]; and VisTex uses visual exemplars to prompt a vision-language detector [33]. No Time to Train uses reference-to-query correspondences in a training-free reference-based segmentation/detection pipeline [2], while FSOD-VFM combines proposal generation, segmentation, and graph-based score refinement [6].

Detection confidence calibration. Classification calibration measures how well confidence agrees with class correctness [10]. Detection calibration also accounts for box localization and which detections are evaluated. Prior work studies both detector calibration and its measurement. Examples include multivariate calibration and D-ECE [14], localization-aware measures [21], and density-based estimation of calibration error [23]. Calibration based on box size [8] or IoU-based modeling of duplicate detections [9] can improve confidence reliability and detection performance. Kuzucu et al. [15] show that detection thresholds and evaluation metrics can affect comparisons between methods. PTPC adjusts confidence scores using region-specific evidence, which can reorder detections and change AP. A strictly increasing calibration map, however, preserves their order within each class and leaves AP unchanged for the same detections. We therefore assess PTPC's effect on detection ranking separately from how reliably its confidence scores reflect correctness.

3. Methods

For each detection labeled as a novel class, PTPC compares its region features with support prototypes and updates its confidence score (Fig. 1). The contextual module measures compatibility with the predicted label relative to other novel classes. The Guard compares the region's closest novel-class prototype with its closest base-class prototype. Full PTPC applies these modules in sequence, while compatible detectors can use the Guard alone.

3.1. Problem Setting

Following the standard few-shot object detection formulation [4, 13, 24, 29], categories are partitioned into disjoint base and novel sets C_b and C_n. The base data $\mathcal{D}_{\text{base}}$ contain abundant annotations, while $\mathcal{D}_{\text{novel}}$ provides K labeled support examples per novel class.

Let D denote a few-shot detector whose parameters remain fixed during rescoring. For a query image $I \in [0,1]^{H_I \times W_I \times 3}$, its post-NMS detection set is

$$\mathcal{Y}_D(I) = \left\{(b_i, c_i, s_i^d)\right\}_{i=1}^{N_I}, \tag{1}$$

where b_i is an axis-aligned box in pixel coordinates, $c_i \in C_b \cup C_n$ is the predicted class, and $s_i^d \in [0,1]$ is the native confidence score.

Our goal is to refine the scores of novel-labelled detections while preserving every box, class label, and detection identity. Base-labelled detections retain their original scores: $\hat{s}_i = s_i^d$ for $c_i \in C_b$.

3.2. Frozen Encoder for Full PTPC

PTPC uses a frozen foundation-model encoder ϕ whose full-image forward pass exposes a spatial grid of patch tokens. We take the normalized final-block spatial tokens, exclude prefix tokens, and apply each checkpoint's native color conversion and channel normalization. Let S_ϕ denote the encoder's square input resolution and let $R(I)$ include this preprocessing and the deterministic square resize used by the implementation. The encoder maps the resized image to a grid of patch tokens,

$$\phi : \mathbb{R}^{S_\phi \times S_\phi \times 3} \to \mathbb{R}^{H \times W \times d}, \qquad Z = \phi(R(I)), \tag{2}$$

where H, W are the token-grid height and width, d the embedding dimension, and $Z_{u,v} \in \mathbb{R}^d$ the token at grid row u and column v. Flattening the grid gives $T(I) = \{(p_r, t_r)\}_{r=1}^{HW}$ with $t_r = Z_{u,v}$ at flat index $r = (u-1)W + v$ and patch-center coordinate $p_r = (x_r, y_r) \in [0, S_\phi]^2$ given by

$$x_r = \frac{v - \frac{1}{2}}{W} S_\phi, \qquad y_r = \frac{u - \frac{1}{2}}{H} S_\phi, \tag{3}$$

placing each token at the center of its grid cell.

3.3. Contextual Novel-Prototype Module

Token selection inside each detector box. Let Π be the deterministic rescaling map from the original image frame to the $S_\phi \times S_\phi$ encoder frame:

$$\tilde{b}_i = \Pi(b_i) = (\tilde{x}_i^1, \tilde{y}_i^1, \tilde{x}_i^2, \tilde{y}_i^2),$$
$$\tilde{x}_i^j = S_\phi \frac{x_i^j}{W_I}, \qquad \tilde{y}_i^j = S_\phi \frac{y_i^j}{H_I}, \qquad j \in \{1,2\}, \tag{4}$$

with center $p_{b_i} = (x_{b_i}, y_{b_i}) = \big(\frac{1}{2}(\tilde{x}_i^1 + \tilde{x}_i^2),\ \frac{1}{2}(\tilde{y}_i^1 + \tilde{y}_i^2)\big)$. Using the token centers $p_r = (x_r, y_r)$ from Eq. (3), let $\Omega_{\text{in}}(b_i) = \{r \mid \tilde{x}_i^1 \le x_r \le \tilde{x}_i^2,\ \tilde{y}_i^1 \le y_r \le \tilde{y}_i^2\}$ collect the token indices $r \in \{1, \dots, HW\}$ whose centers fall inside the rescaled box. PTPC then selects

$$\Omega(b_i) = \begin{cases} \Omega_{\text{in}}(b_i), & \Omega_{\text{in}}(b_i) \neq \emptyset, \\ \big\{\arg\min_r \|p_r - p_{b_i}\|_2\big\}, & \text{otherwise}, \end{cases} \tag{5}$$

falling back to the nearest token whenever no center lies inside the rescaled box so that $\Omega(b_i)$ is never empty.

Center-weighted RoI pooling. Gaussian pooling emphasizes box-center tokens while retaining their full-image context. For box width w_i, height h_i, and token stride $\delta = \min(S_\phi/W, S_\phi/H)$, set $\sigma_{x,i} = \max(w_i/2, \delta)$, $\sigma_{y,i} = \max(h_i/2, \delta)$, and $d_{r,i} = \big((x_r - x_{b_i})/\sigma_{x,i}\big)^2 + \big((y_r - y_{b_i})/\sigma_{y,i}\big)^2$; with $\text{norm}(z) = z/\|z\|_2$, the descriptor is

$$q_p(I, b_i) = \text{norm}\Big(\textstyle\sum_{r \in \Omega(b_i)} a_r(b_i)\ \text{norm}(t_r)\Big),$$
$$a_r(b_i) = e^{-d_{r,i}/2} \Big/ \textstyle\sum_{u \in \Omega(b_i)} e^{-d_{u,i}/2}. \tag{6}$$

Novel patch supports and prototypes. For each novel class $c \in C_n$ with K-shot support set $\mathcal{S}_c = \{(I_c^j, b_c^j)\}_{j=1}^K$, applying the same extractor (Eqs. (4)–(6)) to every support pair and averaging over the K shots gives the class patch prototype $\mu_c^p = \text{norm}\big(\frac{1}{K}\sum_j q_p(I_c^j, b_c^j)\big)$.

Competitive compatibility over novel classes. Given a query detection (b_i, c_i), PTPC compares the patch-level query descriptor with all novel patch prototypes by competitive scoring over the novel label space:

$$s_p(b_i, c) = \frac{\exp\big(q_p(I, b_i)^\top \mu_c^p / \tau\big)}{\sum_{c' \in C_n} \exp\big(q_p(I, b_i)^\top \mu_{c'}^p / \tau\big)}. \tag{7}$$

The inner products are cosine similarities because the descriptors and prototypes are normalized; $\tau > 0$ controls the sharpness of the comparison. The predicted label receives less weight when competing novel prototypes also match the region well. We evaluate $s_p(b_i, c_i)$ at the detector's existing label. This score measures relative compatibility within C_n, rather than the probability of a correct detection: base classes and background are absent from the denominator, and its scale depends on $|C_n|$.

Fusion with detector confidence. PTPC fuses this novel-prototype evidence with the original detector confidence:

$$s_{\text{fuse}}(b_i) = w_d\, s_i^d + w_\phi\, s_p(b_i, c_i), \qquad w_d + w_\phi = 1, \tag{8}$$

with $w_d, w_\phi \in [0,1]$: the frozen-encoder branch is a corrective signal driven by the novel support set, not a replacement detector. Following the common PCB evaluation interface, the novel module evaluates Eq. (8) only where the native score satisfies $\ell_N < s_i^d \le u_N$; outside this band we set $s_{\text{fuse}}(b_i) = s_i^d$.

3.4. Base-Confusability Guard Module

A base-class object can receive high novel compatibility when one novel prototype fits better than the others. The Guard adds a base-class reference. The binary-limit Guard-descriptor ablation gives similar mean AP50 with crop-CLS and patch-RoI features (supplement).

Using the detector's region features. Some foundation-model detectors already compute a region feature $h_D(I, b_i)$ for every final detection. In that case we set $q_g^D(I, b_i) = \text{norm}(h_D(I, b_i))$. The host adapter must expose the same semantic feature space for annotated support regions. Query and support localization mechanisms need not be identical—for example, a host may expose mask-pooled query features but only box-derived support masks—and each such choice is reported.

Crop-level guard descriptor. Let $\text{crop}(I, b_i) \in [0,1]^{S \times S \times 3}$ be the region inside b_i, cropped from I and resized to the encoder crop resolution $S \times S$, and let ϕ_g be the frozen encoder's native global crop representation: its CLS embedding when a prefix token is defined, and its pre-logit global pool otherwise (as for prefix-free SigLIP models). It shares weights with ϕ and differs only in input (a crop at resolution S) and output (one global embedding instead of the spatial patch grid). The crop-level descriptor is $q_g(I, b_i) = \text{norm}(\phi_g(\text{crop}(I, b_i))) \in \mathbb{R}^d$. For compactness, write $\bar{q}_g = q_g$ for full PTPC and $\bar{q}_g = q_g^D$ for a native-host adapter.

Novel and cached base region prototypes. Applying $\bar{q}_g$ to the novel support set $\mathcal{S}_c$ and averaging over the K shots yields the novel region prototype μ_c^g; the same descriptor applied to M_k retained crops of each base class $k \in C_b$ yields a cached base prototype ν_k^g, used only as negative

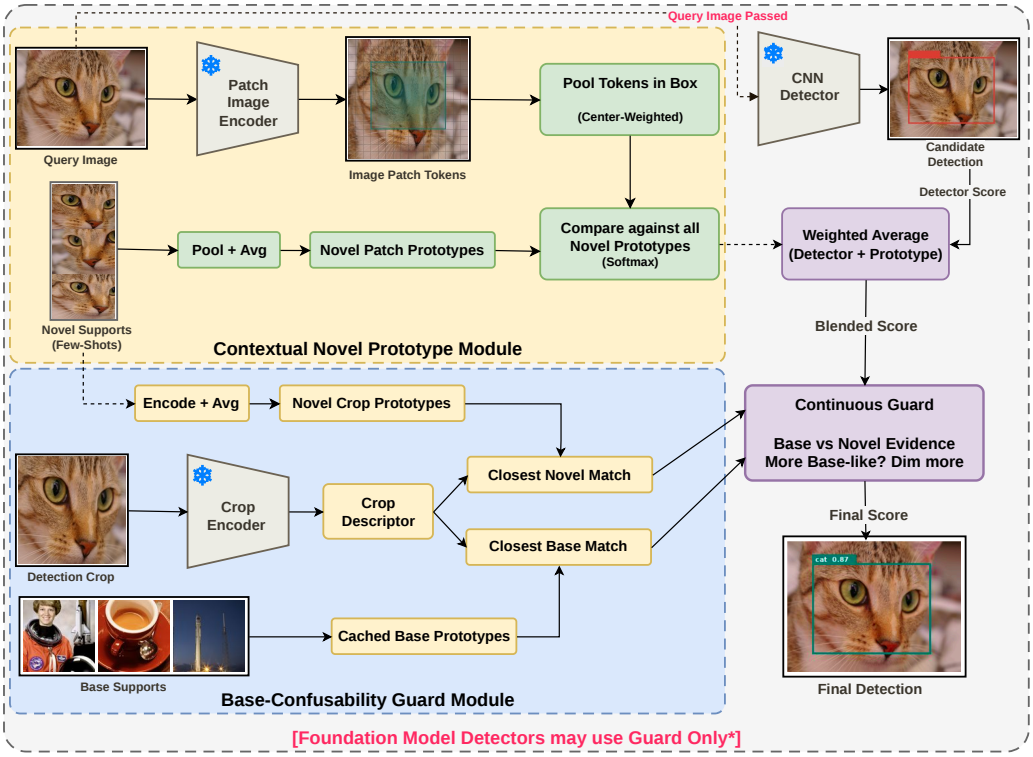


Figure 1. **PTPC overview.** The novel-prototype module pools patch features inside each box into one descriptor and compares it with novel-class prototypes. The Guard uses the relative strength of base and novel matches to reduce the confidence score. Full PTPC uses both modules, while compatible foundation-model detectors can use the Guard on its own.

evidence:

$$\mu_c^g = \mathrm{norm}\Big(\tfrac{1}{K}\textstyle\sum_j \bar{q}_g(I_c^j, b_c^j)\Big),$$
$$\nu_k^g = \mathrm{norm}\Big(\tfrac{1}{M_k}\textstyle\sum_m \bar{q}_g(I_k^m, b_k^m)\Big). \tag{9}$$

Multiplicative score suppression. The Guard compares $\bar{q}_g(I, b_i)$ against the nearest novel and nearest base region prototype, $\delta_i^+ = \max_{c\in C_n} \bar{q}_g(I,b_i)^\top \mu_c^g$ and $\delta_i^- = \max_{k\in C_b} \bar{q}_g(I,b_i)^\top \nu_k^g$. The maximum over C_n deliberately asks whether the region resembles the *novel set*. Label-specific novel competition is the role of the novel-prototype module; the supplementary reference-set audit separately tests using only $\mu_{c_i}^g$ as the positive reference. Because maxima depend on the number of candidates, the same analysis also varies the number of retained base classes; changing crops per prototype M_k is a separate source of variation.

The input s_i^{in} is $s_{\mathrm{fuse}}(b_i)$ for full PTPC and s_i^d for native Guard-only use. Eligibility uses the Boolean mask $B_i = \mathbf{1}[\ell_G < s_i^{\mathrm{in}} \le u_G]$, not a slice of a presumed score-sorted prefix. In full PTPC this mask is recomputed after the novel module. For eligible detections, the Guard smoothly reduces confidence as the strongest base-class match exceeds the strongest novel-class match. The margin m sets the midpoint of this transition. With σ the logistic sigmoid and margin temperature $\eta > 0$,

$$s_{\mathrm{final}}(b_i) = s_i^{\mathrm{in}}\left[1 - (1-\gamma)B_i\,\sigma\left(\frac{\delta_i^- - \delta_i^+ - m}{\eta}\right)\right], \tag{10}$$

where $0 \le \ell_G < u_G \le 1$ and $\gamma \in (0, 1]$. For eligible detections, the multiplier approaches one when the base–novel similarity gap is far below m, and γ when it is far above m; η controls the transition width. For example, with $s_i^{\mathrm{in}} = 0.8$ and $\gamma = 0.3$, an eligible detection at $\delta_i^- - \delta_i^+ = m$ receives a multiplier of 0.65, giving a final score of 0.52. Increasing the gap to $m + 2\eta$ lowers the score smoothly to approximately 0.31. The binary-limit control replaces $\sigma((\delta_i^- - \delta_i^+ - m)/\eta)$ with $\mathbf{1}[\delta_i^- - \delta_i^+ > m]$, corresponding to $\eta \downarrow 0$ away from the boundary. Unless explicitly marked as a binary limit, "Guard" and "PTPC" refer to the continuous rule.

Caching and query-time cost. Support prototypes are computed once. The deployed full crop-CLS path caches $(2|C_n| + |C_b|)d$ scalars; Guard-only or a patch-RoI composition that shares the novel supports caches $(|C_n| + |C_b|)d$. Query inference uses one full-image pass for the novel module; a crop-CLS Guard additionally encodes its eligible

crops. Native Guard-only transfer instead reuses host descriptors and adds only prototype-similarity matrix products, with no new query encoder pass. Exact complexity, cache accounting, latency, and hardware are supplementary.

Rescoring and probability calibration. Region-dependent rescoring can change within-class order and AP. For a final score $s \in (0, 1)$, temperature scaling

$$p_T(s) = \sigma\left(\frac{\text{logit}(s)}{T_{\text{cal}}}\right), \qquad T_{\text{cal}} > 0. \tag{11}$$

is strictly increasing: $T_{\text{cal}} = .5$ maps .30/.70 to approximately .155/.845 without changing order. It can improve ECE or Brier while preserving AP on the same population, without new filtering, truncation, or quantization ties. The region-specific temperatures τ and η can change rank and differ from T_{cal}. We report final-score calibration and its fitting-label budget separately.

4. Experiments

4.1. Experimental Setup

Benchmarks and paired protocol. We evaluate full PTPC with DeFRCN [24] and MFDC [32] on PASCAL VOC [3] and MS COCO [19] under the SNIDA protocol [30]; FSCE [27] is a supplementary detector-family control. Guard-only transfer uses FSOD-VFM [6] and No-Time-to-Train [2]. Our CNN rows use the authors' code and canonical seed-0 supports. Every paired comparison shares the checkpoint, support, post-NMS boxes, labels, detection identities, count, and order; only scores change.

Frozen configuration and label accounting. Full PTPC uses DINOv2-B/14, $(w_d, w_\phi, \tau) = (.4, .6, .1)$, $(m, \gamma) = (.05, .3)$, score bands $(.05, 1]$, and $M = 20$ cached base crops per class. Its non-η tuple is selected once on labelled VOC Split 1 development data and then frozen across detectors, other splits, and COCO. For native-host transfer, *every* Guard parameter is selected on the other benchmark; target labels only form the normal K-shot prototypes. Auxiliary diagnostics use a test-disjoint 1,000-image labelled population and are excluded from strict-K-shot claims. Complete tuples and grids, preprocessing, label exclusions, identity hashes, and probability-map fitting are specified in the supplement.

Metrics. We report AP, AP50, and AP75. Same-class one-to-one matching at IoU$\geq .5$ defines correctness for 15-bin ECE [10] and Brier score. Class-balanced LaECE$_0$ [15, 21], threshold sweeps, calibration baselines, and full definitions are supplementary.

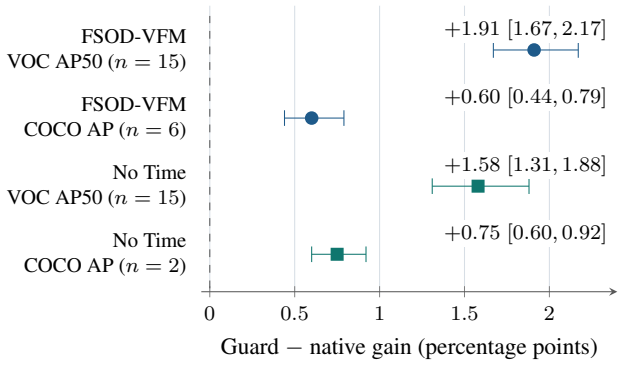


Figure 2. **Paired uncertainty of the canonical continuous Guard.** Mean gains with marginal 95% intervals from 2,000 paired image resamples, conditional on fixed checkpoints, supports, and source-selected parameters. Shared image draws preserve dependence across settings; n counts settings.

4.2. Main Results

PASCAL VOC. On PASCAL VOC (Table 1), canonical PTPC-B improves the identical DeFRCN outputs from 49.6 to 61.4 mean AP50 and MFDC from 57.7 to 63.3, improving all 30 paired detector–shot cells. Gains persist at 10-shot, so they are not explained only by stabilizing a one-shot prototype; the smaller MFDC gain instead tracks its lower remaining ranking headroom. On the native hosts, FSOD-VFM rises 78.2→80.1 and No Time rises 68.1→69.7 mean AP50, each improving all 15 cells. Published retraining and proposal-replacement methods provide context.

MS COCO. On MS COCO (Table 2), canonical PTPC-B improves its paired DeFRCN detector from 15.7 to 18.1 mean AP (+2.4) and MFDC from 16.4 to 18.8 (+2.5), improving all 12 paired cells. The smaller one-shot gains are consistent with noisier novel prototypes, yet the improvement remains broadly stable from two shots onward rather than vanishing with K. FSOD-VFM rises 39.8→40.4 mean AP; No Time rises 36.3→37.1 at 10 shots and 37.0→37.7 at 30 shots. These gains show that the relative scoring cues transfer across datasets and native hosts. Methods that replace the output set can also change proposals or labels.

4.3. Ablations and Scope

Paired uncertainty. Mean-gain intervals exclude zero for all four detector–dataset combinations (Fig. 2). Three individual-setting intervals include zero, so the evidence is stronger for the average improvement than for every setting. Protocols and per-setting intervals are supplementary.

Repeated fine-tunes. The positive paired gains in Table 3 indicate stability across optimizer seeds with fixed supports.

Table 1. **PASCAL VOC novel AP50.** Published rows use their original protocols; our paired rows share seed-0 detector outputs. Guard rows rescore the corresponding *our impl.* host with COCO-selected $\eta = .04$ and no extra VOC labels. †Our No-Time reproduction uses box-derived, not offline-SAM, support masks. Binary/encoder controls are supplementary. **Bold**/underlined: best/second best per column.

Method	Novel Split 1					Novel Split 2					Novel Split 3					Mean
	1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
Norm-VAE [34], CVPR 2023	62.1	64.9	67.8	69.2	67.5	39.9	46.8	54.4	54.2	53.6	58.2	60.3	61.0	64.0	65.5	59.3
DeFRCN [24], ICCV 2021 (our impl.)	48.9	56.3	54.5	60.4	56.1	28.4	39.5	47.5	52.1	47.0	43.7	47.2	51.2	54.6	56.9	49.6
MFDC [32], ECCV 2022 (our impl.)	61.7	66.1	60.9	68.8	66.5	41.3	41.1	45.5	54.8	52.9	58.0	60.7	59.1	63.0	64.5	57.7
SNIDA-MFDC [30], CVPR 2024	64.9	67.9	69.7	71.4	70.5	42.2	47.8	54.5	56.6	54.9	58.1	61.3	60.7	63.6	66.0	60.7
MPAD [28], ICLR 2025	69.1	69.5	69.6	69.9	68.9	58.4	59.7	61.8	61.8	63.5	**70.1**	69.8	69.9	70.4	71.4	66.9
ViT / Foundation-Model methods																
DE-ViT-B [38], CoRL 2024	56.9	61.8	68.0	73.9	72.8	45.3	47.3	58.2	59.8	60.6	58.6	62.3	62.7	64.6	67.8	61.4
FM-FSOD-L [11], CVPR 2024	40.1	53.5	57.0	68.6	72.0	33.1	36.3	48.8	54.8	64.7	39.2	50.2	55.7	63.4	68.1	53.7
VisTex-GLIP [33], ICCV 2025	69.4	71.8	73.4	74.6	74.8	66.5	70.1	73.1	74.2	74.4	68.9	70.2	71.1	72.3	72.6	71.8
No Time to Train [2], arXiv 2025	70.8	72.3	73.3	77.2	79.1	54.5	67.0	76.3	75.9	78.2	61.1	67.9	71.3	70.8	72.6	71.2
No Time to Train (our impl.)†	71.9	70.8	64.0	69.9	76.5	54.6	62.9	71.6	72.7	74.1	57.7	65.2	68.4	69.3	71.5	68.1
FSOD-VFM [6], ICLR 2026	77.5	82.3	83.0	85.8	85.8	64.8	68.0	77.4	79.5	81.6	65.3	75.1	78.7	78.2	79.3	77.5
(our impl.)	77.9	83.0	84.0	86.6	86.5	65.4	69.2	78.5	79.8	82.2	65.5	75.5	79.7	79.0	80.3	78.2
Ours (DeFRCN + PTPC-B)	64.7	69.7	70.5	73.6	71.8	38.0	46.4	58.0	60.6	60.1	56.3	58.6	61.6	65.1	66.4	61.4
Ours (MFDC + PTPC-B)	70.2	67.4	73.6	75.9	76.4	42.0	43.6	56.0	60.1	60.5	60.8	63.3	64.2	67.2	68.4	63.3
Ours (No Time + Guard)†	73.8	73.3	67.0	73.2	79.4	55.0	63.7	72.8	73.9	75.1	59.0	66.9	69.1	70.3	72.2	69.7
Ours (FSOD-VFM + Guard)	**81.5**	**84.6**	**85.4**	**88.3**	**88.3**	**67.6**	**73.2**	**81.8**	**82.3**	**84.0**	67.3	**76.5**	**80.4**	**79.8**	**81.0**	**80.1**

Table 2. **MS COCO novel AP.** Published protocols are unpaired; our rows share seed-0 outputs. Guard rows use VOC-selected $\eta = .01/.005$ for FSOD-VFM/No Time and no extra COCO labels. †Box-derived-mask No-Time reproduction; lower shots are nested support prefixes. **Bold**/underlined: best/second best per column.

Method	1	2	3	5	10	30	Mean
DeFRCN [24], ICCV 2021 (our impl.)	9.3	12.9	14.8	16.1	18.5	22.6	15.7
MFDC [32], ECCV 2022 (our impl.)	10.8	13.9	15.0	16.4	19.4	22.7	16.4
SNIDA-MFDC [30], CVPR 2024	12.0	15.4	16.4	17.8	20.7	23.8	17.7
OWOBJ-DeFRCN [37], CVPR 2025	11.9	15.6	17.9	19.4	23.8	26.4	19.2
MPAD-DeFRCN [28], ICLR 2025	18.3	18.5	18.8	18.9	–	–	–
ViT / Foundation-Model methods							
DE-ViT-L [38], CoRL 2024	–	–	–	–	34.0	34.0	–
FM-FSOD-L [11], CVPR 2024	5.7	11.0	15.7	21.9	27.7	37.0	19.8
VisTex-GLIP [33], ICCV 2025	**47.9**	**51.8**	**52.2**	**52.6**	**52.7**	**53.6**	**51.8**
No Time to Train [2], arXiv 2025	–	–	–	–	36.6	36.8	–
No Time to Train (our impl.)†	–	–	–	–	36.3	37.0	–
FSOD-VFM [6], ICLR 2026	–	–	–	–	44.0	45.8	–
(our impl.)	27.0	35.0	40.4	43.6	45.5	47.3	39.8
AgentDet-Qwen3-8b [17], CVPR 2026	9.4	10.8	16.1	23.9	31.6	37.5	21.6
Ours (DeFRCN + PTPC-B)	10.5	15.7	17.4	18.6	21.1	25.1	18.1
Ours (MFDC + PTPC-B)	12.5	17.0	17.7	19.6	21.7	24.6	18.8
Ours (No Time + Guard)†	–	–	–	–	37.1	37.7	–
Ours (FSOD-VFM + Guard)	27.1	35.2	41.1	44.6	46.4	48.0	40.4

These are binary-limit controls, rather than repeats of the canonical continuous Guard.

Alternative support draws. Table 4 shows positive binary-limit gains across all 120 VOC pairs, with greater variability at low shots. Four MFDC COCO 10-shot draws likewise give 2.5 ± 0.4 AP, with all four gains positive.

Continuous Guard and rule selection. On FSOD-VFM outputs, the continuous Guard exceeds its binary limit by 0.28 VOC AP50 and 0.11 COCO AP, wins 14/21 AP and 18/21 AP50 comparisons against the binary limit, and improves over native in all 21 settings. The direction persists for DeFRCN/MFDC (VOC +0.672/+0.596; COCO +0.0816/+0.0898) and No-Time (+0.39 VOC AP50, +0.04

Table 3. **Repeated fine-tunes.** DeFRCN VOC 5-shot AP50 with binary-limit PTPC-B over five fine-tunes per split. Δ is the paired gain; summaries show mean±SD.

Split	Repeat	DeFRCN	+ PTPC-B	Paired Δ
1	0	62.14	75.21	+13.07
	1	62.58	76.03	+13.45
	2	61.44	75.13	+13.69
	3	62.11	75.50	+13.38
	4	61.86	75.06	+13.20
	mean	$62.03_{\pm 0.42}$	$75.39_{\pm 0.39}$	$+13.36_{\pm 0.24}$
2	0	50.47	60.22	+9.76
	1	50.26	59.64	+9.39
	2	50.07	59.65	+9.58
	3	51.13	60.62	+9.49
	4	50.84	60.32	+9.48
	mean	$50.55_{\pm 0.43}$	$60.09_{\pm 0.43}$	$+9.54_{\pm 0.14}$
3	0	56.87	67.48	+10.62
	1	57.68	67.77	+10.09
	2	57.27	67.43	+10.17
	3	57.43	67.54	+10.11
	4	57.66	67.88	+10.23
	mean	$57.38_{\pm 0.33}$	$67.62_{\pm 0.19}$	$+10.24_{\pm 0.22}$
All splits		$56.65_{\pm 0.27}$	$67.70_{\pm 0.19}$	$+11.05_{\pm 0.09}$

Table 4. **Alternative support draws.** Binary-limit PTPC-B minus detector AP50 over four official support draws, mean±SD.

		1-shot	2-shot	3-shot	5-shot	10-shot
DeFRCN	S1	$+11.8_{\pm 3.5}$	$+8.9_{\pm 0.6}$	$+9.9_{\pm 0.9}$	$+9.4_{\pm 0.8}$	$+9.7_{\pm 0.6}$
	S2	$+5.3_{\pm 1.1}$	$+7.1_{\pm 1.0}$	$+7.7_{\pm 0.7}$	$+7.7_{\pm 0.5}$	$+8.0_{\pm 0.8}$
	S3	$+13.6_{\pm 3.9}$	$+9.9_{\pm 1.1}$	$+10.5_{\pm 2.6}$	$+9.9_{\pm 1.6}$	$+9.6_{\pm 1.0}$
MFDC	S1	$+6.9_{\pm 1.0}$	$+5.2_{\pm 0.8}$	$+6.9_{\pm 1.1}$	$+7.4_{\pm 0.9}$	$+5.4_{\pm 0.9}$
	S2	$+4.8_{\pm 0.4}$	$+3.4_{\pm 1.6}$	$+7.4_{\pm 3.2}$	$+6.1_{\pm 1.8}$	$+5.4_{\pm 0.5}$
	S3	$+5.7_{\pm 1.8}$	$+4.4_{\pm 1.0}$	$+5.2_{\pm 0.7}$	$+6.3_{\pm 3.0}$	$+4.8_{\pm 1.1}$

COCO AP). Thus continuity modestly refines the main gain.

Mechanism controls. Absolute-only rules remain at native AP50 (78.19/78.26 versus 78.23), while the binary-limit relative Guard reaches 79.75. Temperature fitting reduces native ECE from 0.150 to 0.040 and binary-limit Guard ECE/Brier to 0.0117/0.0064 without changing AP. Full auxiliary-label diagnostics are supplementary.

Encoder robustness. All eight encoders in Table 5 exceed the MFDC baseline, including CLIP and SigLIP, while DINOv2-L is strongest on this split. The non-monotonic DINOv2 scale trend argues against capacity alone explaining the gain; all-split VOC and COCO sweeps are supplementary.

Robustness and deployment controls. Support budget saturates: $M = 5/20/50$ gives 65.42/66.18/66.34 AP50, whereas base coverage $B = 5/10/15$ gives 63.33/64.41/66.18. Detector/novel/full inference takes 48.5/78.4/112.4 ms per image; further controls are supplementary.

Table 5. **Frozen-encoder robustness on MFDC VOC Split 1.** Novel AP50; only the encoder changes. Each encoder uses its deployed input resolution and patch size.

Foundation model	1	2	3	5	10	Mean
MFDC (no PTPC)	61.70	66.10	60.90	68.80	66.50	64.80
DINOv2 [22]						
DINOv2-S/14	67.27	61.43	71.27	74.88	74.97	69.96
DINOv2-B/14	68.70	**71.10**	71.10	74.80	73.60	71.86
DINOv2-L/14	**69.06**	68.37	**72.97**	**76.19**	**76.93**	**72.70**
DINOv2-G/14	68.45	59.81	70.97	74.78	75.64	69.93
CLIP [25]						
CLIP-B/16	65.69	59.16	68.06	71.78	71.00	67.14
CLIP-L/14	64.29	62.99	68.20	74.86	72.94	68.66
SigLIP [36]						
SigLIP-B/16	59.95	63.58	68.29	74.53	75.40	68.35
SigLIP-L/16	59.44	54.11	66.05	72.67	74.66	65.39

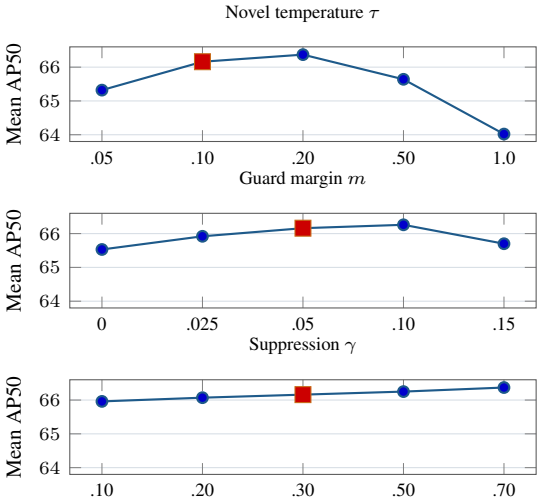


Figure 3. **Hyperparameter sensitivity on DeFRCN VOC 5-shot.** Three-split mean AP50 for one-at-a-time sweeps. Squares mark the Split 1-selected settings; split-wise values are supplementary.

Component ablation. Table 6 separates encoder, pooling, competition, and Guard. DINOv2 improves crop-CLS AP50 (56.09→60.68) but worsens ECE (0.217→0.369). Changing the encoder alone does not reproduce the full PTPC result. Competition plus Guard loses 1.62 AP50 with global crops but gains 2.15 with localized patch-RoI evidence; the complete patch configuration is best in mean AP50 and Brier.

These are binary-limit controls. Native-host experiments omit novel competition, isolating the Guard on host descriptors.

Table 6. **DeFRCN VOC 5-shot factorial ablation.** All rows use the same supports; results for each split are novel AP50. **Bold**/underlined: best/second best.

Encoder	Pooling	Comp.	Guard	Split 1	Split 2	Split 3	Mean AP50	ECE↓	Brier↓
—	—	—	—	49.98	52.68	50.31	50.99	0.151	0.059
R101	Crop-CLS	—	—	60.59	52.96	54.74	56.09	0.217	0.084
R101	Crop-CLS	✓	—	60.20	48.57	54.59	54.45	0.228	0.103
R101	Crop-CLS	—	✓	61.90	54.01	57.68	57.86	**0.114**	0.044
R101	Crop-CLS	✓	✓	62.04	51.63	57.90	57.19	0.123	0.053
DINOv2-B	Crop-CLS	—	—	65.06	56.54	60.43	60.68	0.369	0.158
DINOv2-B	Crop-CLS	✓	—	68.80	59.51	61.61	63.31	0.205	0.073
DINOv2-B	Crop-CLS	—	✓	70.23	59.76	63.75	64.58	0.166	0.054
DINOv2-B	Crop-CLS	✓	✓	68.94	58.49	61.46	62.96	0.205	0.073
DINOv2-B	Patch-RoI	—	—	65.95	56.77	60.82	61.14	0.381	0.168
DINOv2-B	Patch-RoI	✓	—	68.05	57.26	60.00	61.77	0.240	0.102
DINOv2-B	Patch-RoI	—	✓	72.11	59.05	65.14	65.43	0.173	0.058
DINOv2-B	Patch-RoI	✓	✓	**74.82**	**60.31**	**67.60**	**67.58**	**0.114**	**0.043**

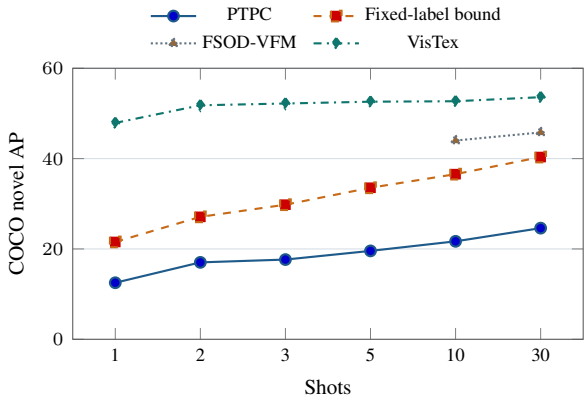


Figure 4. **Fixed-output ceiling.** The bound exceeds PTPC but remains below methods that can change detector outputs. COCO novel AP.

Fixed-output ceiling. At 10-shot, PTPC (21.69 AP) remains 14.87 points below the fixed-label bound (Fig. 4). Yet the 10/30-shot bounds (36.56/40.35) remain below output-changing FSOD-VFM (44.0/45.8) [6] and VisTex (52.7/53.6) [33], despite remaining ranking headroom.

Guard selectivity. Across 533,937 checks, the crop Guard suppresses 79.9% of detections without novel-ground-truth overlap versus 10.6% with overlap (patch-RoI: 90.7%/9.8%). It acts more often on detections without novel-object overlap.

Failure cases and threshold effects. Of 40,583 changed Split 2 5-shot detections, 29,839 are demoted false positives and 9,320 are promoted. At threshold 0.5, however, 2,255 false positives cross upward versus 1,222 downward, so AP and a chosen operating point can disagree. The audit also finds 3,179 poorly localized boxes and 1,139 incorrect labels. Increasing margin 0.05→0.15 restores horse recall but costs 0.46 mean AP50.

Hyperparameter sensitivity. One-at-a-time sweeps (Fig. 3) span 2.35 AP50 for τ, 0.73 for m, and 0.42 for γ. Alternatives improve the Split 1-selected mean by at most 0.21; uniform pooling changes it by 0.03.

5. Conclusion

PTPC improves detection ranking through comparisons with competing novel and base prototypes. Gains are largest on the CNN baselines. On FSOD-VFM and our No-Time reproduction, the standalone Guard provides smaller improvements, with paired bootstrap intervals supporting positive mean gains on both datasets.

Limitations. PTPC cannot fix missed objects, poor boxes, or wrong labels; confusable classes can trade threshold recall for precision. The continuous Guard does not consistently improve calibration over its binary limit. Native transfer needs compatible descriptors and representative base support and is tested only on FSOD-VFM and our box-mask No-Time reproduction. Auxiliary 1,000-image studies are not strict K-shot results, and budget/class-count controls use deterministic subsets. Deployment therefore requires host-specific validation of the Guard and probability map.

Code Availability

The implementation is available at https://github.com/Trex1102/ptpc-code.

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